Homework Assignment 2

Exploring patterns of environmental justice and historical redlining in LA

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```
# Load libraries
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(knitr)
library(kableExtra)
```

```
# Load data
ejscreen <- st_read("data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb")
mi_la <- st_read('data/mapping-inequality/mapping-inequality-los-angeles.json')
mi_nh <- st_read('data/mapping-inequality/mapping-inequality-new-haven.json')
switrs <- read_csv('data/SWITRS_LA/Crashes.csv')|>
  as_tibble() |>
  drop_na(POINT_X) |>
  drop_na(POINT_Y) |>
  st_as_sf(coords = c("POINT_X", "POINT_Y"), crs = 4326, remove = FALSE) |>
  st_transform(crs = st_crs(3857))
```

```
# Filter to California
california <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "CA")

# Filter to San Joaquin County
los_angeles <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))

# Find the average values for all variables within counties
california_counties <- aggregate(california, by = list(california$CNTY_NAME), FUN = mean)
```

```r
# Check CRS
st_crs(ejscreen) #ID["EPSG",3857]]
st_crs(mi_la) #ID["EPSG",4326]]
st_crs(switrs) # Changed when loading data

# Transform CRS to match EJS Screen
mi_la <- st_transform(mi_la, crs= st_crs(los_angeles))

# Check if CRS match between 2 datasets
if(st_crs(los_angeles) == st_crs(switrs)) {
  print("CRS matches")
} else {
  warning("CRS does not match")
}
```

```r
# Data Exploration
head(mi_la)
```

```
Simple feature collection with 6 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -13187810 ymin: 4042916 xmax: -13143940 ymax: 4054200
Projected CRS: WGS 84 / Pseudo-Mercator
  area_id city_id grade    fill label name category_id sheets         area
1    7761      16     A #76a865    A1                1      1 3.359915e-04
2    7775      16     A #76a865   A10                1      1 1.814147e-04
3    7808      16     A #76a865   A11                1      1 5.978184e-05
4    8025      16     A #76a865   A12                1      1 2.587288e-04
5    7608      16     A #76a865   A13                1      1 1.326238e-04
6    7797      16     A #76a865   A14                1      1 2.629402e-04
                                                                                bound
1                   [ [ 34.136969999999998, -118.46807 ], [ 34.153350000000003, -
118.42031 ] ]
2                               [ [ 34.1691, -118.11198 ], [ 34.188940000000002, -
118.0979 ] ]
3                   [ [ 34.155970000000003, -118.11301 ], [ 34.163069999999998, -
118.09853 ] ]
4 [ [ 34.122999999999998, -118.18331000000001 ], [ 34.159080000000003, -
118.15949999999999 ] ]
5          [ [ 34.127760000000002, -118.18619 ], [ 34.146239999999999, -
118.16952000000001 ] ]
```

```
6 [ [ 34.105049999999999, -118.12802000000001 ], [ 34.130420000000001, -
118.07402999999999 ] ]
  residential commercial industrial     label_coords
1        TRUE      FALSE      FALSE 34.147, -118.452
2        TRUE      FALSE      FALSE 34.177, -118.104
3        TRUE      FALSE      FALSE 34.159, -118.102
4        TRUE      FALSE      FALSE 34.148, -118.171
5        TRUE      FALSE      FALSE 34.133, -118.175
6        TRUE      FALSE      FALSE 34.119, -118.105
                        geometry
1 MULTIPOLYGON (((-13186622 4...
2 MULTIPOLYGON (((-13148113 4...
3 MULTIPOLYGON (((-13148279 4...
4 MULTIPOLYGON (((-13154761 4...
5 MULTIPOLYGON (((-13155134 4...
6 MULTIPOLYGON (((-13149259 4...
```

```r
names(mi_la)
```

```
 [1] "area_id"      "city_id"      "grade"        "fill"         "label"
 [6] "name"         "category_id"  "sheets"       "area"         "bounds"
[11] "residential"  "commercial"   "industrial"   "label_coords" "geometry"
```

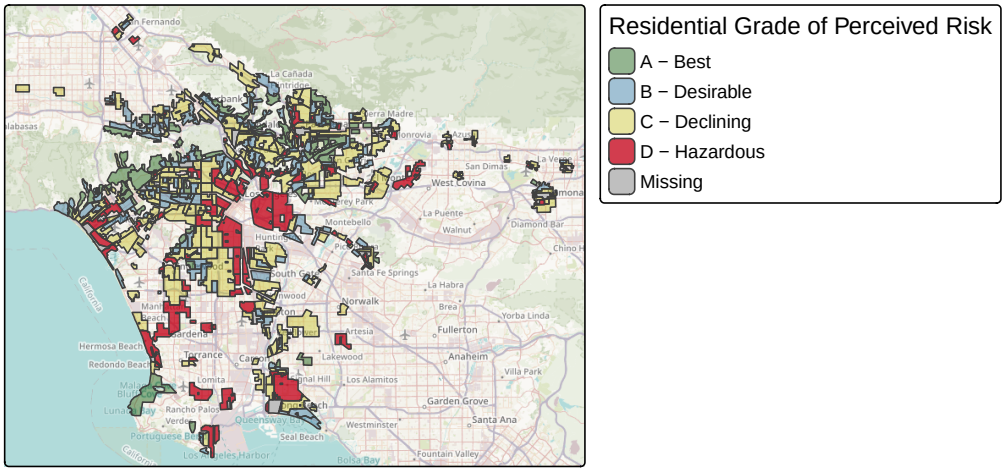
Part 1: Legacy of redlining in current environmental (in)justice

```r
# Create specific colors per HOLC grades for all visualizations
grade_colors_transparent <- c("A" = "#0855016f", "B" = "#00588b5e", "C" = "#bfb90360", "D" ="

grade_colors<- c("A" = "#76A865", "B" = "#7CB5BD", "C" = "#FFFF00", "D" ="#e30c25")

# Visualize historical HOLC grades in Los Angeles
tm_basemap("OpenStreetMap.HOT",alpha = 0.7) +
tm_shape(mi_la) +
tm_polygons(
    fill= "grade",
    title = "Residential Grade of Perceived Risk",
    palette = grade_colors_transparent,
    labels = c("A - Best", "B - Desirable", "C - Declining", "D - Hazardous")
    )+
tm_layout(scale = 0.8)+
  tm_title("Historical Redlining in Los Angeles", size = 1.8)
```

Historical Redlining in Los Angeles




```
# Match the coordinate refrence
la_ejs_join <- st_join(los_angeles,mi_la) |>
    st_drop_geometry()

# Classify all NAs as "No Grade"
ejscreen_redlining_percent <- la_ejs_join |>
  mutate(
    grade = ifelse(is.na(grade), "No Grade", grade)) |>
  count(grade) |>
  mutate(percent = (n / sum(n)) * 100)

# Render the table
kbl(ejscreen_redlining_percent, digits = 2,
    col.names = c("HOLC Grade", "Count", "Percentage"),
    caption = "Census Block Groups by HOLC Grade") |>
  kable_styling(bootstrap_options = "striped")
```

```
# Filter data for wanted columns
new_df <- la_ejs_join |>
     select(ID, grade, PM25, LOWINCPCT,LIFEEXPPCT, P_LIFEEXPPCT)

# Calculate mean of PM25 and group by grade
```

Table 1: Census Block Groups by HOLC Grade

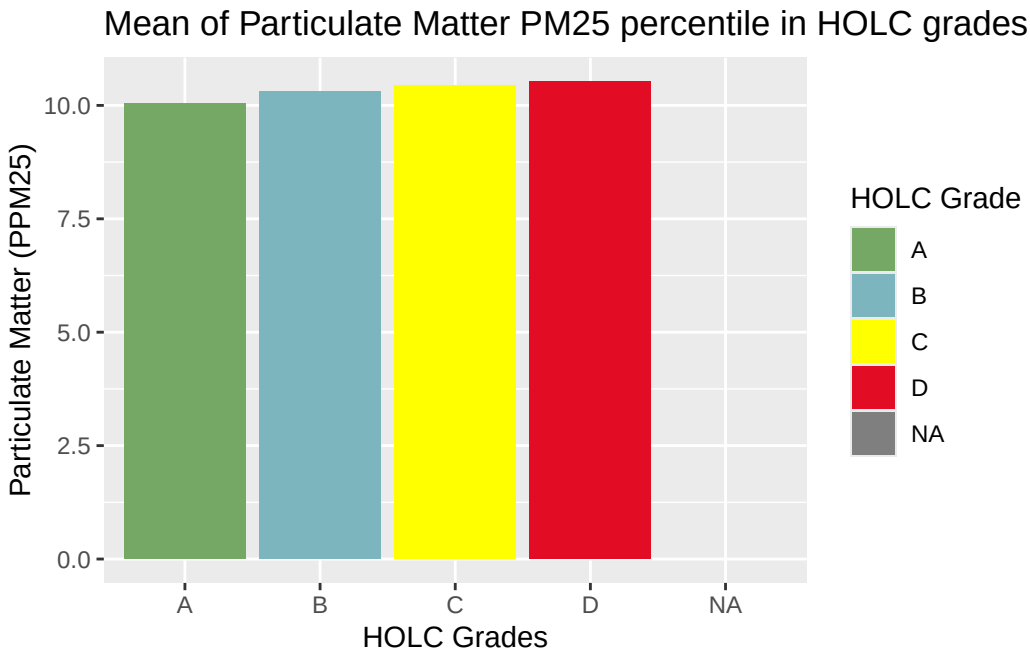
HOLC Grade	Count	Percentage
A	449	5.00
B	1239	13.79
C	3058	34.02
D	1346	14.98
No Grade	2896	32.22

```
PM25_grade<- new_df |>
    summarise(PM25_mean = mean(PM25), .by= grade)

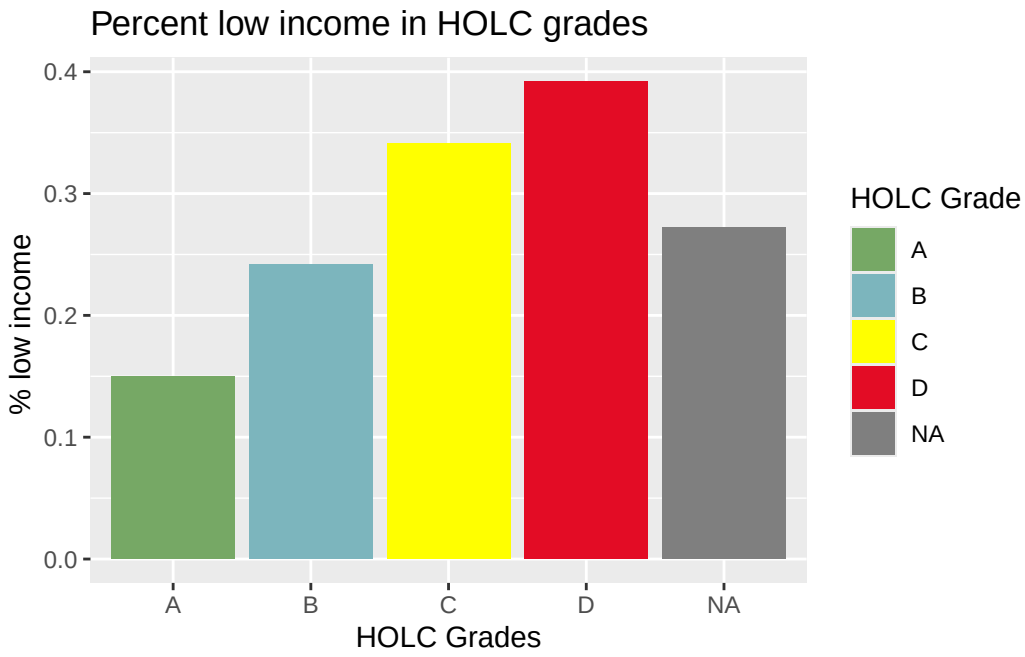
# Calculate mean of percent low income and group by grade
LOWINCPCT_grade<- new_df  |>
    summarise(LOWINCPCT_mean = mean(LOWINCPCT), .by= grade)

# Calculate mean of percentile for low life expectancy and group by grade
LIFEEXPPCT_grade<- new_df |>
  group_by(grade) |>
  summarise(LIFEEXPPCT = mean(LIFEEXPPCT, na.rm = TRUE))
```

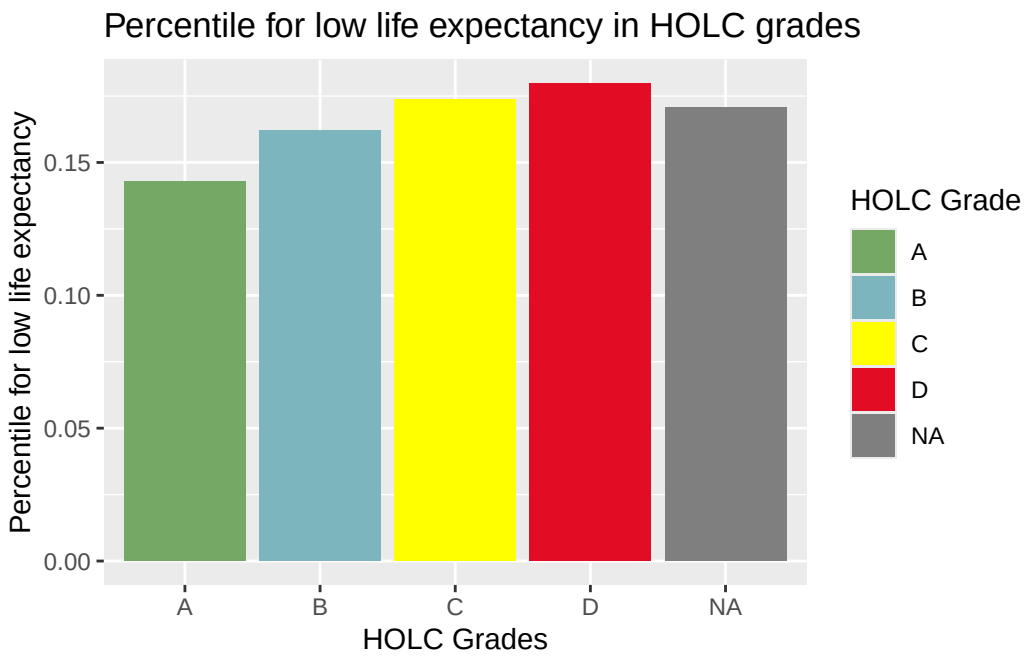
```
# Visualize PM25 means grouped by HOLC grades
ggplot(data = PM25_grade, aes(fill = grade)) +
    geom_col(mapping = aes(x = grade,y= PM25_mean))+
    scale_fill_manual(values = grade_colors, name = "HOLC Grade")+
  labs(
    title = "Mean of Particulate Matter PM25 percentile in HOLC grades",
    x = "HOLC Grades",
    y = "Particulate Matter (PPM25)"
  )
```




```
# Visualize low income means grouped by HOLC grades
ggplot(data = LOWINCPCT_grade, aes(fill = grade)) +
    geom_col(mapping = aes(x = grade,y=LOWINCPCT_mean ))+
    scale_fill_manual(values = grade_colors, name = "HOLC Grade")+
  labs(
    title = "Percent low income in HOLC grades",
    x = "HOLC Grades",
    y = "% low income "
  )
```




```
# Visualize low life expectancy grouped by HOLC grades
ggplot(data = LIFEEXPPCT_grade,aes(fill = grade)) +
    geom_col(mapping = aes(x = grade,y=LIFEEXPPCT))+
    scale_fill_manual(values = grade_colors, name = "HOLC Grade")+
  labs(
    title = "Percentile for low life expectancy in HOLC grades",
    x = "HOLC Grades",
    y = "Percentile for low life expectancy"
  )
```



Reflection

Analyzing past systematic marginalization offers insight into how these moments in history have shaped, and continue to shape, present-day life. All of the visualizations summarizing EJScreen data on percent low income, PM25 percentile, and low life expectancy percentile show the highest values in grade D neighborhoods and lowest in grade A. This simple analysis points to the lasting impacts of historical marginalization, but it doesn't fully capture the impact of redlining on Los Angeles County, since most of the current county area was never assigned an HOLC grade.

Part 2: Legacy of redlining in pedstrian and bike accidents

```
# A figure summarizing the percent of observations within redlined neighborhoods within each

# Filter to two columns I want
select_crashes <- switrs |>
  select(POINT_X, POINT_Y, PEDESTRIAN_ACCIDENT, BICYCLE_ACCIDENT, NUMBER_KILLED)

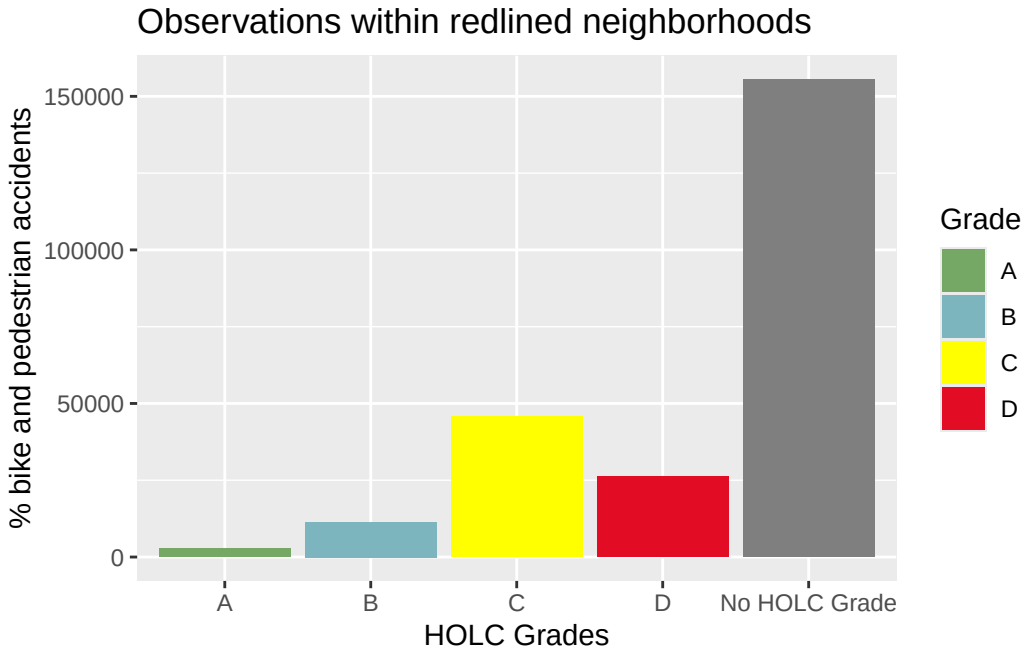
# Drop na or change them under a different name?
select_crashes <- select_crashes |>
```

```
  mutate(PEDESTRIAN_ACCIDENT = ifelse(is.na(PEDESTRIAN_ACCIDENT), "No accident", PEDESTRIAN_A
  mutate(BICYCLE_ACCIDENT = ifelse(is.na(BICYCLE_ACCIDENT), "No accident", BICYCLE_ACCIDENT))

# Join datasets/ drop geometry
join_crashes_redlining <- st_join(select_crashes, mi_la) |>
  st_drop_geometry()

# Create a percent table
crashes_redlining_percent <- join_crashes_redlining |>
  mutate(
    grade = ifelse(is.na(grade), "No HOLC Grade", grade)
  ) |>
  count(grade) |>
  mutate(percent = (n / sum(n)) * 100)
```

```
# Visualize the percentage of bike and pedestrian crashes and fatalities within each HOLC gra
ggplot(data = crashes_redlining_percent,aes(fill = grade)) +
    geom_col(mapping = aes(x = grade,y= n))+
    scale_fill_manual(values = grade_colors, name = "Grade")+
  labs(
    title = "Observations within redlined neighborhoods",
    x = "HOLC Grades",
    y = "% bike and pedestrian accidents"
  )
```



Reflection

Bike and pedestrian accidents do not follow the pattern in the previous analysis, grade C areas have the highest percentage, followed by grade D areas. Assigning HOLC grades to areas with no assigned HOLC grade could help determine how much of this pattern is tied to redlining.